

Apriori-Based Analysis of Cybersecurity Attack Patterns

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Abstract

Cybersecurity network traffic contains complex combinations of protocols, services, connection states, and attack behaviours. This study investigates the use of the Apriori association-rule mining algorithm to discover interpretable patterns associated with cybersecurity attack categories in the UNSW-NB15 intrusion-detection dataset. The analysis focuses on categorical network characteristics—protocol, service, and connection state—and transforms each selected network record into a transaction suitable for frequent-itemset mining. A minimum support of 0.005 and minimum confidence of 0.30 were used to generate candidate association rules. Rules containing attack categories in their antecedents were removed, and redundant variants were reduced to obtain eight representative attack patterns. The

resulting rules were evaluated on the testing partition using the same preprocessing procedure. The strongest relative association was Protocol=unas $\rightarrow$ DoS, with testing lift of 9.5969. The rule Service=pop3 + State=FIN $\rightarrow$ Exploits achieved 93.81% testing confidence and 7.5754 testing lift, while Service=dns + State=INT $\rightarrow$ Generic achieved 99.44% testing confidence and 4.1147 lift. In contrast, the representative Fuzzers pattern showed weaker generalization. The findings demonstrate that Apriori can support interpretable exploration of cybersecurity attack patterns, while also showing the importance of independent validation and careful interpretation of association strength.

Keywords: *Cybersecurity, Association Rule Mining, Apriori, UNSW-NB15, Intrusion Detection, Attack Patterns, Frequent Itemsets, Network Traffic*

1. Introduction

Modern network environments generate large volumes of traffic containing heterogeneous characteristics. Identifying relationships between network conditions and malicious activity is therefore an important component of cybersecurity analytics. Conventional intrusion detection research often emphasizes classification and prediction; however, interpretable descriptions of recurring attack patterns can also be valuable to analysts. Association rule mining provides a natural framework for this purpose because it expresses relationships in human-readable form as item combinations and their associated outcomes.

Apriori is a classical frequent-itemset mining algorithm introduced by Agrawal and Srikant for discovering association rules in transactional databases [1]. In cybersecurity, a network record can be represented as a transaction containing categorical items such as a protocol, service, and connection state. Mining these transactions can reveal combinations that occur disproportionately often with labelled attack categories.

This study applies Apriori to the UNSW-NB15 dataset, a benchmark network intrusion-detection dataset containing normal traffic and nine attack types. The official UNSW source provides separate training and testing CSV partitions, enabling the discovered associations to be evaluated beyond the data used for mining [2]. The objective is not to build a predictive classifier or infer causality, but to identify and validate descriptive associations between selected network characteristics and attack categories.

2. Dataset and Experimental Design

2.1 Dataset

The UNSW-NB15 dataset was developed at UNSW Canberra and contains modern normal activities and synthetic contemporary attack behaviours. The official dataset description reports 2,540,044 total records and 49 generated features with a class label. The supplied training partition contains 175,341 records and the testing partition contains 82,332 records [2]. The attack taxonomy includes Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and

Worms, in addition to the normal class.

Property	Value
Training records	175,341
Testing records	82,332
Total records in official dataset	2,540,044
Features in official dataset	49 + class label
Attack types	9
Classes represented in analysis	Normal + 9 attack categories

Table 1. Dataset characteristics used to frame the experiment.

2.2 Feature Selection and Transaction Construction

The Apriori algorithm operates on item-based transactions rather than raw continuous measurements. Accordingly, four categorical fields were selected for the transaction representation: proto, service, state, and attack_cat. To reduce very sparse categories, commonly occurring values were retained. The protocol values were tcp, udp, unas, arp, ospf, and sctp; the service values were -, dns, http, smtp, ftp-data, ftp, ssh, and pop3; and the state values were INT, FIN, CON, and REQ.

After this filtering step, 161,567 of the 175,341 training records remained. Each retained record was converted into a transaction containing items such as Protocol=tcp, Service=dns, and State=INT, together with the corresponding attack category. The same categorical filtering procedure was applied to the testing data, leaving 77,910 records for validation.

2.3 Association-Rule Mining

Frequent itemsets were generated using Apriori with minimum support of 0.005 (0.5%). Association rules were then generated using minimum confidence of 0.30 (30%). Candidate rules were restricted to those whose consequent contained an attack category, while attack items were excluded from antecedents. Rules predicting the normal class were excluded from the final attack-pattern analysis.

For an association rule $X \rightarrow Y$, support measures the proportion of transactions containing both X and Y, while confidence measures the proportion of transactions containing X that also contain Y. Lift compares the observed co-occurrence with the expected co-occurrence under an independence assumption. A lift greater than 1 indicates positive association. These metrics are interpreted jointly rather than treating any single metric as sufficient evidence of a strong rule.

2.4 Rule Reduction and Validation

The initial mining stage produced a larger set of attack-related rules. Redundant variants with essentially equivalent antecedent information and metrics were removed to produce a compact set of representative rules. The final set contains eight rules covering DoS, Exploits, Generic, and Fuzzers. Each rule was then

evaluated on the testing partition using the same transaction filters. Testing confidence and testing lift were compared with their training counterparts.

3. Results and Discussion

3.1 Final Representative Rules

Attack Pattern	Train Conf.	Test Conf.	Train Lift	Test Lift
DoS Protocol=unas	33.76%	33.17%	7.0772	9.5969
Exploits Service=pop3 + State=FIN	94.27%	93.81%	5.5809	7.5754
Exploits Service=smtp	61.54%	56.67%	3.6430	4.5764
Generic Service=dns + State=INT	99.20%	99.44%	4.0226	4.1147
Generic Protocol=udp + Service=dns	82.78%	85.06%	3.3568	3.5194
Exploits Protocol=ospf + State=INT	49.60%	41.22%	2.9366	3.3287
Generic Protocol=udp + State=INT	76.17%	74.21%	3.0887	3.0708
Fuzzers Protocol=udp+Service=-+State=INT	37.48%	16.77%	3.4583	2.4095

Table 2. Final representative association rules and their training/testing metrics.

3.2 DoS Association Pattern

The strongest relative association was Protocol=unas → DoS. The rule had training confidence of 33.76% and testing confidence of 33.17%, a change of only −0.59 percentage points. Its lift increased from 7.0772 in training to 9.5969 in testing. Thus, the protocol condition remained strongly associated with DoS relative to the DoS prevalence in the corresponding filtered testing data. The result should not be interpreted to mean that every record using this protocol is a DoS attack; confidence remains substantially below 100%.

3.3 Exploits Association Patterns

The strongest high-confidence rule was Service=pop3 + State=FIN → Exploits. Training confidence was 94.27%, compared with 93.81% in testing, while lift increased from 5.5809 to 7.5754. The very small confidence change suggests strong consistency across the two partitions. The Service=smtp → Exploits rule showed a moderate decrease in confidence from 61.54% to 56.67%, while its lift increased from 3.6430 to 4.5764. Similarly, Protocol=ospf + State=INT → Exploits decreased from 49.60% to 41.22% confidence but retained a testing lift of 3.3287. These results illustrate why confidence and lift should be examined together.

3.4 Generic Association Patterns

The Generic category produced highly stable patterns. Service=dns + State=INT → Generic achieved 99.20% training confidence and 99.44% testing confidence. Its lift increased from 4.0226 to 4.1147. The simpler combination Protocol=udp + Service=dns → Generic also generalized well, with testing confidence of 85.06% and testing lift of 3.5194. Protocol=udp + State=INT → Generic achieved 74.21% testing confidence and 3.0708 testing lift. Together, these rules indicate consistent associations involving DNS, UDP, and the INT state in the analysed data.

3.5 Fuzzers Pattern and Generalization

The representative Fuzzers rule, Protocol=udp + Service=- + State=INT → Fuzzers, showed the largest reduction in confidence. Confidence declined from 37.48% in training to 16.77% in testing, while lift decreased from 3.4583 to 2.4095. Although the testing lift remained above 1, the substantial change indicates weaker generalization than the strongest rules. This finding is important because it demonstrates that a rule satisfying the mining thresholds on training data may still have substantially different association strength on unseen data.

3.6 Training–Testing Comparison

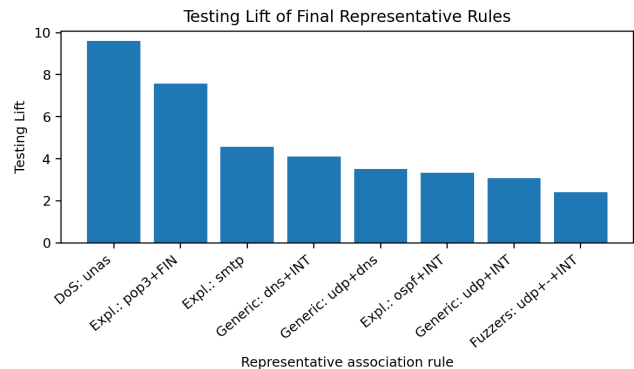


Figure 1. Testing lift for the eight final representative rules.

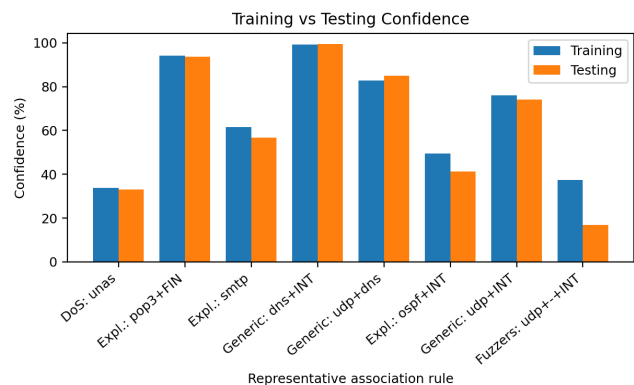


Figure 2. Training and testing confidence for the eight final representative rules.

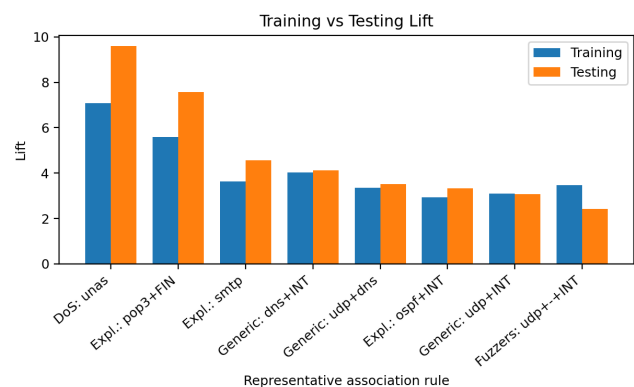


Figure 3. Training and testing lift for the eight final representative rules.

Across the final rules, several patterns show close training/testing confidence values, particularly the POP3–FIN Exploits and DNS–INT Generic

associations. The DoS rule shows a small confidence change while its relative lift increases in testing. The Fuzzers rule is the main exception, with a pronounced decline in both confidence and lift. The comparison therefore supports the use of testing validation as an essential component of association pattern analysis.

3.7 Interpretation of Lift and Confidence

The results reinforce the complementary roles of confidence and lift. The DoS rule has the highest testing lift (9.5969) but only 33.17% testing confidence. In contrast, the POP3–FIN Exploits rule has 93.81% testing confidence and 7.5754 testing lift. A high lift indicates that an event is much more common under the antecedent than would be expected from its overall prevalence, whereas confidence describes how frequently the consequent occurs when the antecedent occurs. Consequently, the two measures answer different questions and should not be conflated.

4. Limitations

4.1 Dependence on categorical features

The transaction representation focuses on protocol, service, and connection state. Continuous network characteristics are not directly represented, so some information in numerical traffic features may be missed.

4.2 Threshold sensitivity

A minimum support of 0.005 and minimum confidence of 0.30 were selected for this study. Different thresholds can produce different frequent itemsets and rules, and patterns below the support threshold are not represented.

4.3 Rare attack categories

The final representative rules are concentrated on DoS, Exploits, Generic, and Fuzzers. Less frequent categories did not produce final representative rules under the selected criteria, limiting their coverage.

4.4 Dataset-specific behaviour

The findings are based on UNSW-NB15. Testing on its official testing partition provides within-dataset validation, but broader generalization requires evaluation on independent intrusion-detection datasets.

4.5 Unequal generalization

The Fuzzers rule showed a large confidence decline between training and testing. This indicates that some discovered associations can be sensitive to data distribution.

4.6 Association is not causation or prediction

The rules are descriptive statistical associations with labelled attack categories. They do not establish causal relationships and should not be presented as a standalone predictive classifier.

4.7 Computational scalability

Apriori can become computationally demanding as the number of items and transaction complexity increase.

Larger or higher-dimensional cybersecurity datasets may require pruning, optimization, or alternative algorithms.

5. Conclusion

This study investigated Apriori-based association rule mining as an interpretable approach for cybersecurity attack-pattern analysis using the UNSW-NB15 dataset. Selected categorical network characteristics were converted into transactions and mined using a minimum support of 0.005 and minimum confidence of 0.30. Following attack-specific filtering, redundancy reduction, and independent testing validation, eight representative rules were retained.

The results identified several strong associations. Protocol=unas → DoS produced the highest testing lift at 9.5969. Service=pop3 + State=FIN → Exploits produced 93.81% testing confidence and 7.5754 testing lift, while Service=dns + State=INT → Generic achieved 99.44% testing confidence and 4.1147 testing lift. The testing results also revealed that not all patterns generalized equally, with the Fuzzers rule showing a substantial reduction in confidence.

Overall, the findings demonstrate that Apriori can provide compact and interpretable descriptions of recurring cybersecurity patterns. The method is best viewed as a complementary exploratory technique for threat-pattern analysis rather than as a causal or predictive model. The combination of support, confidence, lift, redundancy reduction, and independent validation provides a practical basis for assessing the usefulness and stability of the discovered patterns.

6. Future Scope

6.1 Integrate numerical network features using discretization or hybrid transaction-generation techniques.

6.2 Use adaptive or class-specific support thresholds to improve analysis of rare attack categories.

6.3 Compare Apriori with FP-Growth and Eclat in terms of computational efficiency, pattern quality, and interpretability.

6.4 Validate discovered associations across additional intrusion-detection datasets to assess cross-dataset generalization.

6.5 Develop interactive cybersecurity analytics systems that allow analysts to explore protocol, service, state, and attack associations.

6.6 Investigate incremental or streaming association-rule mining for continuously changing network traffic.

6.7 Combine association mining with supervised machine-learning methods to explore interpretable hybrid cybersecurity frameworks.

6.8 Develop automated rule-selection and redundancy-reduction methods to retain compact,

informative pattern sets.

7. References

- [1] R. Agrawal and R. Srikant, "Fast Algorithms for Mining Association Rules in Large Databases," Proceedings of the 20th International Conference on Very Large Data Bases (VLDB), Santiago, Chile, 1994, pp. 487–499.
- [2] N. Moustafa and J. Slay, "UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems (UNSW-NB15 Network Data Set)," 2015 Military Communications and Information Systems Conference (MilCIS), 2015. Dataset information and downloads: <https://research.unsw.edu.au/projects/unsw-nb15-dataset>

Research-data note. Numerical results, final representative rules, training/testing metrics, and figures in this paper are based on the final research-results file supplied for this study. The paper intentionally describes the rules as associations with labelled attack categories rather than as causal or predictive rules.